



AKTU

B.Tech II-Year



Cyber security



Unit-1 in One Shot

Pdf Notes + PYQs + Imp. Questions

Cyber Security

Unit-1 AKTU Syllabus

INTRODUCTION TO CYBER CRIME : Cybercrime- Definition and Origins of the word Cybercrime and Information Security, Who are Cybercriminals? Classifications of Cybercrimes, A Global Perspective on Cybercrimes, **Cybercrime Era**: Survival Mantra for the Netizens. **Cyber offenses**: How Criminals Plan the Attacks, Social Engineering, Cyber stalking, Cybercafe and Cybercrimes, **Botnets**: The Fuel for Cybercrime, Attack Vector.

AKTU PYQs : Unit-1

Q.1	Explain various social engineering techniques used by cyber criminals	AKTU 2024-25
Q.2	Explain the impact of botnets on cybercrime and online security.	AKTU 2024-25
Q.3	Explain how cybercriminals plan cyberattack including the steps involved	AKTU 2024-25
Q.4	Define information security	AKTU 2024-25
Q.5	What are cybercriminals? Classify cybercrime	AKTU 2023-24
Q.6	What is the fuel of cybercrime? How may a criminal plan cyberattack	AKTU 2023-24
Q.7	Explain how cyber crime term originated	AKTU 2023-24
Q.8	Explain different stages of cyberattack	AKTU 2023-24
Q.9	How can botnet activities can be detected and disrupted by cyber security professional	AKTU 2023-24

CYBER CRIME

➤ Cybercrime is a type of crime that is done using computers, the internet, or other digital devices. It involves stealing, harming, or misusing data or systems through technology.

Examples of Cybercrime:

- Hacking into someone's account
- Sending fake emails to steal passwords (phishing)
- Spreading viruses or malware
- Online scams or fraud
- Cyberbullying or online threats

Origin of word cyber crime

Word	Origin
Cyber	From the Greek word "kybernetes" meaning "steering" or "governing". In modern use, "cyber" refers to computers, networks, or digital technology.
Crime	From Latin "crimen" meaning "accusation" or "offense". It refers to illegal acts punishable by law.

- The term "cybercrime" began appearing in the 1990s.
- It became popular as the internet grew and criminals started using computers to commit illegal act

Difference between-

Aspect	Cybercrime	Traditional Crime
Medium Used	Computer, internet, digital devices	Physical tools, weapons, or manual actions
Location	Can happen from any place (remotely)	Happens at a specific physical location
Target	Data, networks, online accounts	People, property, physical objects
Examples	Hacking, phishing, malware, online fraud	Theft, murder, assault, robbery
Evidence Type	Digital (IP logs, emails, digital traces)	Physical (fingerprints, eyewitnesses, CCTV footage)
Speed of Attack	Can happen instantly online	Usually takes time and physical effort
Risk to Criminal	Low (can stay anonymous online)	High (criminal is physically present at the crime scene)

Information Security

Information Security
(also called InfoSec)
means protecting
information from being:

- Accessed by unauthorized people
- Changed without permission
- Deleted or lost

How Information Security is Related to Cybercrime

- Information Security and Cybercrime are closely connected because
- Cybercrime Targets Information, and Information Security Protects It.

Cybercrime	Information Security
Tries to steal, damage, or misuse data	Tries to protect data from such activities
Involves attacks like hacking, phishing, malware	Uses tools like firewalls, encryption, antivirus
Violates confidentiality, integrity, availability	Maintains confidentiality, integrity, availability

Cybercriminals

- **Cybercriminals are individuals or groups who commit crimes using computers, networks, or the internet. They aim to steal data, spread viruses, hack systems, or scam people online.**

TYPES OF CYBERCRIMINALS

Type	Description	Example
Hackers	Break into systems to steal or damage data	Hacking into a bank's server
Crackers	Break security for malicious purposes	Cracking software to remove protection
Phishers	Create fake emails/websites to steal personal info	Fake email pretending to be a bank
Spammers	Send unwanted emails or messages, often with harmful links	Bulk ads or scam links in emails
Scammers/Fraudsters	Trick people online to steal money	Fake online shopping site

...TYPES OF CYBERCRIMINALS

Type	Description	Example
Cyber Terrorists	Use the internet to cause fear or harm to society/government	Attacks on national infrastructure
Cyberstalkers	Harass or threaten people repeatedly using digital platforms	Sending threatening messages on social media
Insiders	Employees who misuse access to company data	Selling customer data from inside a company
Script Kiddies	Inexperienced attackers using ready-made tools	Using downloaded software to launch attacks

Categories of cybercrime

Category	Description
Cybercrime against Individuals	Crimes that target people personally (identity theft, cyberstalking).
Cybercrime against Property	Attacks on data, systems, or networks (hacking, malware, ransomware).
Cybercrime against Government/Society	Attacks aimed at national security or public services (cyber terrorism).

TYPES OF CYBERCRIME

Type of Cybercrime	Category	Description
Hacking	Property	Unauthorized access to systems or data.
Phishing	Individuals	Tricking users to give personal information.
Identity Theft	Individuals	Stealing someone's personal identity or data.
Cyberstalking	Individuals	Online harassment or threats.
Ransomware Attack	Property	Locking data and demanding payment.
Malware Distribution	Property	Spreading harmful software like viruses.

TYPES OF CYBERCRIME

Type of Cybercrime	Category	Description
Online Fraud/Scam	Individuals/Property	Cheating users to steal money or information.
Denial-of-Service (DoS)	Property/Government	Making websites or servers crash.
Cyber Terrorism	Government/Society	Attacks to create fear or disturb national systems.
Cyber Defamation	Individuals	Spreading false info to damage reputation.
Software Piracy	Property	Copying or using software illegally.

Impact of Cybercrime on Individuals, Property, and Government

1. Impact on Individuals

Effect	Description
Financial Loss	Loss of money through online fraud, scams, or identity theft.
Emotional Stress	Anxiety, fear, or trauma due to cyberstalking, harassment, or threats.
Privacy Violation	Personal data (photos, passwords, etc.) being stolen or leaked.
Reputation Damage	Defamation or misuse of personal info on social media.

...Impact of Cybercrime on Individuals, Property, and Government

2. Impact on Property (Data, Devices, Networks)

Effect	Description
Data Theft or Loss	Important files and data stolen, deleted, or encrypted by attackers.
System Damage	Malware or ransomware damages software and systems.
Business Disruption	Online attacks stop business operations or services.
Financial Impact	Costs of fixing damage, recovering data, and improving security.

...Impact of Cybercrime on Individuals, Property, and Government

3.Impact on Government and Society

Effect	Description
National Security Threats	Cyberterrorism or spying attacks can target defense, police, or intelligence.
Public Trust Loss	People lose trust in government services if cyberattacks are frequent.
Critical Infrastructure Hit	Cyberattacks can damage power grids, transport, healthcare systems, etc.
Economic Damage	Loss of GDP due to disrupted services and increased cybersecurity costs.

- Cybercrime has become a worldwide challenge due to the increasing dependence on technology and the internet.
- It affects individuals, businesses, and governments across borders, making it one of the most serious threats in the modern digital world

1. Cybercrime is a Borderless Crime

- Unlike traditional crimes that happen in one country, cybercriminals can attack from any part of the world using the internet.
- A hacker sitting in one country can steal money, data, or personal information from people or systems in another country within seconds.
- Example:
A ransomware attack launched from Russia can lock down computers in the United States or India and demand payment in cryptocurrency.

2. Common Global Cybercrime Threats

Cybercrime	Global Impact
Ransomware	Locks files and demands payment; affects hospitals, banks, and schools.
Phishing & Fraud	Trick users globally to steal money or personal info.
Data Breaches	Leaks millions of users' data from global platforms like Facebook, LinkedIn.
Cyber Terrorism	Attacks on national security systems like power grids or military servers.

3. Why It's Hard to Control Globally

- **Different Laws:** Each country has its own cyber laws, so what is illegal in one place may not be illegal in another.
- **Lack of Cooperation:** Some countries don't share cybercrime information or support investigations.
- **Technical Complexity:** Cybercriminals use encryption, fake identities, and dark web tools to hide.
- **Jurisdiction Issues:** Which country has the right to arrest and punish an attacker who operates internationally?

4. The Need for Global Action

- **Strong international cyber laws**
- **Information sharing between countries**
- **Cybersecurity awareness programs**
- **Training law enforcement globally**

5.International Efforts to Fight Cybercrime

Countries and organizations are trying to work together through:

Organization/Agreement	Purpose
Interpol Cybercrime Unit	Coordinates police efforts worldwide.
Budapest Convention on Cybercrime	First international treaty to fight cybercrime.
UNODC (UN Office on Drugs and Crime)	Helps countries create cybercrime laws and investigate cases.
Europol EC3 (European Cybercrime Centre)	Fights cybercrime in European countries.

Surviving Mantra for Netizens for Online Security

To stay safe online, every netizen (internet user) should follow some key practices. These act as a "survival mantra" to protect personal data, avoid cyber threats, and use the internet responsibly.

1. Use Strong & Unique Passwords

- Use long, complex passwords with a mix of letters, numbers, and symbols.
- Avoid using the same password for all accounts.
- Use a password manager if needed

2. Enable Two-Factor Authentication (2FA)

- Add an extra layer of protection to your accounts using OTPs or authentication apps.
- Even if your password is stolen, your account stays protected.

3. Beware of Phishing Scams

- Don't click on suspicious links or open unknown attachments.
- Check email sender addresses carefully.
- Be alert to fake messages pretending to be from banks or services.

4. Keep Software and Devices Updated

- Regularly update apps, browsers, and operating systems to fix security bugs.
- Turn on automatic updates when possible.

5. Use Antivirus and Firewall

- Install reliable antivirus software and keep it updated.
- Use a firewall to block unauthorized access to your device.

6. Be Cautious on Public Wi-Fi

- Avoid logging into sensitive accounts (banking, email) on public Wi-Fi.
- Use a VPN (Virtual Private Network) if needed.

7. Control What You Share Online

- Don't overshare personal information like location, phone number, or bank details.
- Set your social media accounts to private.

8. Regularly Backup Your Data

- Keep backups of important files on external drives or cloud storage.
- Helps recover your data in case of ransomware or crashes.

9. Log Out After Use

- Always log out of accounts when using shared or public devices.
- Don't save passwords in browsers on public computers.

10. Stay Informed

- Learn about new cyber threats and safety tips.
- Attend awareness sessions, read trusted tech news, or follow cybersecurity channels.

What is a Hacker?

- A hacker is someone who uses technical knowledge to break into computer systems or networks — either for good or bad purposes.

Type	What they do	Purpose
White Hat	Ethical hacker – finds and fixes security flaws.	Legal and helpful
Black Hat	Breaks into systems to steal, damage, or spy.	Illegal and harmful
Grey Hat	Breaks in without permission but doesn't harm.	Illegal but not malicious

What is a Cracker?

- A cracker is someone who breaks the security or copyright protections of software or systems, often to use it illegally.

What Crackers do:

- Remove software copy protection
- Unlock paid features of apps (e.g., cracking a game)
- Break passwords or bypass licensing

What is a Phreaker?

A phreaker is a person who hacks telephone systems to make free calls or explore how phone networks work.

What Phreakers do:

- Bypass telephone billing systems
- Manipulate tone signals to access operator lines
- Often used tricks like blue boxes in old analog systems.

This was more common before mobile phones, but it's part of hacking history.

Cyber Offence

- cyber offence is any illegal act done with the help of a computer, mobile phone, or the internet.
- Just as stealing a bike is an offence in the physical world, stealing data, money, or someone's online identity is an offence in the digital world.

Type of Offence	What Happens	Everyday Example
1. Hacking / Unauthorised Access	Breaking into someone's computer or account.	A teenager guesses a café's Wi-Fi admin password and changes the settings.
2. Data Theft	Copying or stealing files without permission.	An employee emails the company's customer list to a rival firm.
3. Identity Theft	Pretending to be someone else online.	A fraudster uses your Aadhaar details to open a bank account.
4. Phishing & Online Fraud	Tricking people into giving money or passwords.	You get an SMS saying "update KYC or account will freeze" and you enter your OTP on a fake site.

Type of Offence	What Happens	Everyday Example
5. Malware Distribution	Spreading viruses, worms, ransomware, etc.	A free movie download secretly installs ransomware that locks your files for ransom.
6. Denial-of-Service (DoS/DDoS)	Flooding a website so it crashes and no one can use it.	Attackers overload a ticket-booking site during a big sale.
7. Cyber-stalking & Harassment	Repeatedly threatening or following someone online.	Someone keeps sending abusive messages and tracks your social-media activity.
8. Cyber-terrorism	Using the internet to create fear or damage critical infrastructure.	Hacking a city's power-grid control systems to cause a blackout.

How cybercriminal plans attack

Step	What Happens	Example
1. Reconnaissance	Collecting information about the target (organization or person).	Finding an employee's email from LinkedIn to send a fake mail later.
2. Scanning	Scanning systems or websites for weak spots or vulnerabilities.	Using a tool like Nmap to find open ports on a company's server.
3. Gaining Access	Attacking using malware, phishing, or exploiting software flaws.	Sending a fake email with a malware link saying "Your invoice is attached."
4. Maintaining Access	Installing backdoors or creating secret accounts to stay in the system.	Installing a hidden Trojan that lets the attacker re-enter any time.
5. Exploitation	Stealing data, locking files, transferring money, or spying on activities.	Encrypting all files and demanding a ransom (ransomware).
6. Covering Tracks	Deleting logs, hiding malware, or using fake IP addresses to avoid detection.	Using a VPN and deleting system logs after the attack.

Social Engineering

- Social engineering is a type of cyber attack where hackers trick people into giving away confidential information like passwords, OTPs, or bank details by pretending to be someone trustworthy.
- It doesn't attack the computer or software it attacks the human mind.

Why is it called "Social" Engineering?

- Because it uses social skills (like lying, pretending, convincing) to fool people — not code or hacking tools. It manipulates human behavior.

Why Social Engineering is Dangerous?

- **Hard to detect:** It uses human trust.
- **No antivirus can stop it:** Because people are tricked into helping the attacker.
- Even smart people get fooled.



Hacker sends phishing mail



Opens PDF



Malware sends the data to hacker



User receives mail with an attachment



Malware executes & steals sensitive data



Types/techniques of social engineering

Technique	Description	Example
Phishing	Fake emails or websites	You get a fake email from "Amazon" asking to login and update your card info.
Vishing	Voice phishing (phone calls)	A fake call from "tech support" asking for your computer password.
Smishing	SMS-based phishing	You receive an SMS with a fake link: "Click here to win ₹10,000!"
Pretexting	Creating a fake identity	A hacker pretends to be your manager and asks for employee data.
Baiting	Using a physical or digital "bait"	A USB drive labeled "Salary Slips" is dropped near the office, and when someone plugs it in, malware is installed.
Tailgating	Following someone into a secure area	An attacker follows an employee into a building without using access cards.

Cyber-stalking

- Cyber-stalking is when someone uses the internet or technology to repeatedly watch, follow, contact, or harass another person in a way that causes fear or stress.
- Social media (Instagram, Facebook, etc.), Email or chat apps

Why Is Cyber-stalking Dangerous?

- Can cause fear, stress, or depression
- May lead to real-life stalking or violence
- Hard to escape if the stalker hides their identity
- Can damage your reputation, career, or relationships

How to Stay Safe:

- Don't share too much personal info online
- Use strong privacy settings on social media
- Block and report the stalker
- Keep evidence (screenshots, messages)
- Inform police or cybercrime cell

KEY CHARACTERSTICS OF CYBER STALKING

Characteristic	What it Means
Repeated Behavior	Happens again and again, not just once
Unwanted Contact	You don't want to talk, but they keep messaging
Harassment or Threats	Abusive or scary messages, calls, or posts
Surveillance	They keep checking your online activities
Fake Profiles	They use new or false identities to reach you
Emotional Harm	You feel stressed, afraid, or unsafe
Real-World Impact	Can lead to real stalking, harm, or depression

Type	Description
1. Vindictive Cyberstalker	Stalks out of anger or revenge. Their goal is to cause fear, damage reputation, or punish the victim. They may post hateful content or send threats.
2. Composed Cyberstalker	Stalks to annoy, irritate, or disturb the victim, but not necessarily out of hate. Often does it just to feel powerful or superior.
3. Intimate Cyberstalker	A person who wants to form or restore a relationship with the victim (often an ex-lover). Their behavior includes constant messaging, emotional blackmail, or threats.
4. Collective Cyberstalker	Involves a group of people (like online mobs or trolls) working together to target or harass someone online. They spread hate, lies, or threats as a group.
5. Predatory Cyberstalker	Most dangerous type. Tries to gain trust of the victim (often children or teens), then tries to meet them in real life for abuse or harm.
6. Cyberstalker for Extortion	Seeks money, private images, or information by threatening the victim (e.g., sextortion). Usually anonymous and uses blackmail tactics.

Types of cyber stalker

Type	Description
6. Cyberstalker for Extortion	Seeks money, private images, or information by threatening the victim (e.g., Usually anonymous and uses blackmail tactics.
7. Delusional Cyberstalker	Mentally unstable person who believes the victim loves them, even if they've never met. May stalk celebrities or strangers obsessively.

CYBERCAFE

A cyber cafe (also called an internet cafe) is a place where people can pay to use computers with internet access.

Key features:

- Provides internet access, often charged per hour or per minute.
- Offers printing, scanning, or gaming services in some cases.
- Useful for people who don't have internet or a computer at home.

CYBER CAFE ASSOCIATED WITH CYBER CRIME IN INDIA

Anonymity: Criminals use cyber cafes to hide their identity (no personal device or IP address).

Lack of monitoring: Some cafes don't maintain proper user records (ID proofs, usage logs).

Illegal activities: Offenders may use cafes for:

- Hacking or phishing
- Sending threats or fake emails
- Accessing illegal websites

Government action in India:

IT Rules (Cyber Cafe) 2011: Cyber cafes must:

- Verify and record user ID (e.g., Aadhaar, passport)
- Keep usage logs for 1 year
- Maintain CCTV surveillance

some important safety measures individuals should take while using a computer in a cyber cafe:

Before use:

Check the surroundings – Make sure no one is watching you type.

Inspect the computer – Look for any unusual devices (e.g., keyloggers) attached to the keyboard or USB ports.

While using:

- **Avoid online banking or shopping** – Don't enter sensitive information (like card details).
- **Use incognito/private mode** – Prevents saving browsing history and cookies.

While using:

- **Don't save passwords** – Never click "Save Password" in the browser.
- **Log out properly** – From all accounts (email, social media, etc.).
- **Avoid downloading files** – They might contain malware or viruses

After use:

- Clear browser history and cache
- **Delete downloaded files** – From "Downloads" or desktop.
- **Restart the system** – It clears temporary session data.

BOTNETS

A botnet is a group of infected computers or devices that are controlled remotely by a hacker (called a botmaster), without the users' knowledge.

Key Points:

Each infected device is called a bot or zombie.

Botnets are used to:

- Send spam emails
- Steal personal data
- Spread malware

botnet considered as "the fuel of cybercrime"

A botnet is often called "the fuel of cybercrime" because it powers and enables many large-scale cyber attacks.

Reasons why botnets are considered "the fuel of cybercrime":

Massive Power: Thousands or millions of bots act together, making attacks strong and hard to stop.

Anonymity: Hackers hide behind the infected systems, making it hard to trace them.

Versatility:

Botnets are used for many crimes:

- DDoS attacks
- Spamming
- Phishing
- Data theft
- Spreading malware

Low cost, high damage: Criminals can rent botnets cheaply on the dark web to launch powerful attacks.

IMPACT OF BOTNET ON CYBERCRIME

Area	Impact Description
1. DDoS Attacks	Botnets are used to overload websites or servers, causing them to crash (Denial-of-Service).
2. Spamming and Phishing	Botnets send millions of spam emails or phishing links to steal personal data.
3. Credential Theft	Bots can record keystrokes or steal login details (passwords, banking info).
4. Malware Distribution	Used to spread more viruses or ransomware to other systems.
5. Click Fraud	Bots automatically click ads to generate fake revenue.

IMPACT OF BOTNET ON ONLINE SECURITY

Security Threat	How Botnets Affect It
Loss of Control	Users lose control over their devices (used without permission).
Network Slowdown	Infected systems and networks become slow due to hidden bot activities.
Privacy Breach	Bots may record private data like passwords, messages, and documents.
Widespread Damage	A single botnet can attack thousands of devices or websites at once.
Hard to Trace	Cybercriminals hide behind thousands of bots, making it hard for police to catch them.
High Financial Losses	Organizations lose money fixing damage or paying ransoms.

ATTACK VECTOR

- An attack vector is the pathway or method used by a hacker or cybercriminal to gain access to a computer, network, or system to perform malicious activities like stealing data, damaging systems, or spreading malware.

How Attack Vectors Are Launched:

- **Identify Weakness** – Hacker finds a vulnerability (e.g., old software, weak password).
- **Choose Method** – Decides on the type of attack (phishing, malware, etc.).
- **Launch Attack** – Sends email, infects system, or breaks in using the chosen vector.
- **Gain Access** – Gets control or steals data.
- **Cause Harm** – Deletes files, steals info, or demands ransom.

Phishing

Fake emails or messages that look real but trick users into clicking malicious links or giving passwords.

A fake email from "your bank" asking for login details.

Malware

Software designed to harm your system, like viruses, worms, spyware, or ransom-ware.

Downloading a free game that secretly installs a virus.

**Social
Engineering**

Tricking people into revealing confidential information by pretending to be trustworthy.

A caller pretends to be from tech support to get your password.

**Brute-force
attacks**

Repeatedly trying passwords until the correct one is found.

A hacker uses a program to guess your Facebook password.

Attack Vector	Explanation	Example
Unpatched Software	Using outdated software with known security flaws that haven't been fixed.	A hacker attacks a system running old Windows software.
Drive-by Downloads	Visiting a compromised website automatically installs malware on your device.	You visit a hacked website and malware downloads silently.
Insider Threats	Employees or others within the organization misusing their access.	A disgruntled worker steals company data and sells it.

Professionals Detect Botnet Activities

. Network Traffic Analysis

They monitor unusual network behavior, such as: High outbound traffic

Repeated communication with unknown IPs or domains
(Command & Control servers)

Tool Examples: Wireshark, Zeek, Suricata

2. Intrusion Detection Systems (IDS)

IDS tools scan for suspicious patterns, like known botnet signatures or attack techniques.

They alert when a device matches known malware behavior.

Tool Examples: Snort, OSSEC, Suricata

3. Behavioral Analysis

Monitoring system behavior (CPU spikes, auto-emailing, file changes).

Sudden changes like mass emailing or accessing files repeatedly can indicate a bot.

Example: A home PC sending 10,000 emails per hour → likely part of a spam botnet.

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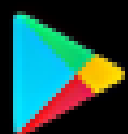


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Empowering Learners, Transforming Futures



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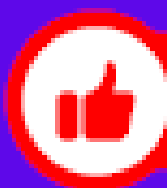


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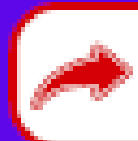
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